Eidgenössische Technische Hochschule Zürich
Swiss Federal Institute of Technology Zurich

ENGINEERING DESIGN AND COMPUTING

Structural Optimization 2026

Project #1 Report

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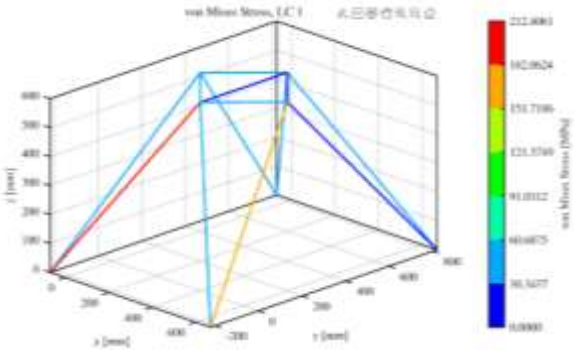
ID Number: 20-823-639

Task 1

After setting up everything as defined in the task presentation and the template of the function EDACFEM, the objective function is computed, and we obtain the following result:
At x = [0 0], f = 5196.741215
The optimal solution is found at x* = [-67.873447, -220.395326] and f *= 4585.106056

Task 2

Iteration	Func-count	f(x)	Step-size	First-order optimality
0	3	5196.74		7.33
1	12	4763.98	12.4226	2.7
2	15	4633.23	1	1.23
3	18	4589.67	1	0.36
4	21	4585.28	1	0.0731
5	24	4585.11	1	0.0153
6	27	4585.11	1	0.0094
7	30	4585.11	1	0.00174
8	33	4585.11	1	0.000274
9	36	4585.11	1	1.8e-05
10	39	4585.11	1	1.94e-06



Optimization completed: The first-order optimality measure, 2.328476e-07, is less than options.OptimalityTolerance = 1.000000e-06.

- Number of optimization steps: **10** - Max displacement: 1.0934 mm -Min displacement: 0.0000 mm
- Number of function evaluations: **39** - Max stress: 212.4061 MPa - Min stress: 0.5713 MPa

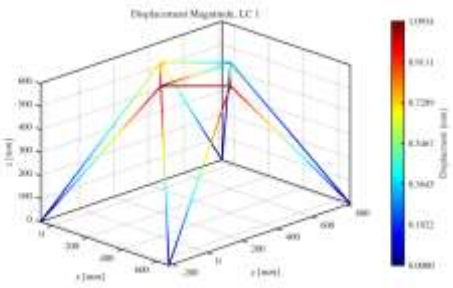
-Optimal design x* = [-67.873447, -220.395326] - Optimal objective f(x*) = 4585.106056

Task 3

Total number of iterations: 17, x* = -67.873591 -220.395112 - f(x*) = 4585.106056 - Total number of f evaluations: 241
-Max displacement: 1.0934 mm - Min displacement: 0.0000 mm -Max stress: 212.4061 MPa - Min stress: 0.5713 MPa
To respect the condition of the gradient the maximum number of iteration is 30

Task 4

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Iteration 0: x = [0.000000, 0.000000], f = 5196.741215, ||grad|| = 7.341678e+00
Iteration 1: x = [-4.455232, -65.924730], f = 4844.645535, ||grad|| = 3.740901e+00, alpha = 9.000000e+00
Iteration 2: x = [-10.231231, -132.572393], f = 4673.756584, ||grad|| = 1.854706e+00, alpha = 1.000000e+00
Iteration 3: x = [-15.835749, -182.508468], f = 4621.084060, ||grad|| = 1.256250e+00, alpha = 1.000000e+00
Iteration 4: x = [-19.974340, -208.186744], f = 4611.310440, ||grad|| = 1.183140e+00, alpha = 1.000000e+00
Iteration 5: x = [-22.868359, -217.885191], f = 4608.893106, ||grad|| = 1.152367e+00, alpha = 1.000000e+00
Iteration 6: x = [-28.266398, -227.809375], f = 4605.371952, ||grad|| = 1.066652e+00, alpha = 1.000000e+00
Iteration 7: x = [-39.253147, -238.020878], f = 4599.133375, ||grad|| = 8.594021e-01, alpha = 1.000000e+00
Iteration 8: x = [-54.565411, -241.012483], f = 4591.534216, ||grad|| = 5.374155e-01, alpha = 1.000000e+00
Iteration 9: x = [-66.960587, -232.773943], f = 4586.361862, ||grad|| = 2.089199e-01, alpha = 1.000000e+00
Iteration 10: x = [-69.821175, -223.527398], f = 4585.196838, ||grad|| = 4.908965e-02, alpha = 1.000000e+00
Iteration 11: x = [-68.680302, -220.828150], f = 4585.112858, ||grad|| = 1.540377e-02, alpha = 1.000000e+00
Iteration 12: x = [-67.956736, -220.363448], f = 4585.106149, ||grad|| = 2.082586e-03, alpha = 1.000000e+00
Iteration 13: x = [-67.875536, -220.300390], f = 4585.106056, ||grad|| = 1.096796e-04, alpha = 1.000000e+00
Iteration 14: x = [-67.873979, -220.395745], f = 4585.106056, ||grad|| = 2.264931e-06, alpha = 1.000000e+00
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Optimal design x* = [-67.873979, -220.395745] Optimal objective f(x*) = 4585.106056
Number of optimization steps = 14 Number of function evaluations = 48
m Max displacement: 1.0934 mm Min displacement: 0.0000 mm
Max stress: 212.4061 MPa Min stress: 0.5714 MPa